# Sefik EGRIBOZ (M.Sc., PMP, PSPO, ITIL)



## SOLUTIONS ARCHITECT

**Airline Operations | Schedule Management & Distribution**
**Operations Control (IOCC) | Flight Operations & Flight Planning**

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**Immediately available | Open to relocation | sefikegriboz.net**

## PROFESSIONAL SUMMARY

**Key results:** **98%** reduction in reported ATC confusion events with AI in production | restricted-airspace dispatches down from **18** a year to **zero** since go-live | **EUR 8M** cumulative savings

Airline operations solutions architect with 20+ years inside the systems an operations control center (OCC) runs on: movement control, aircraft rotation/tail assignment, IATA Type B MVT/LDM, ACARS, NOTAMs and crew management.

Joined the first production implementation of the **Lufthansa Systems NetLine suite, then owned it through fleet growth** from **60 to 500+ aircraft** and 10M to 85M passengers a year as customer-side product owner for NetLine, SchedConnect and flight planning across three carriers: a five-vendor RFP, POC evaluations, technical due diligence and key-user training.

**IATA schedule data standards in production**, not on paper: SSIM Chapters 4 (SSM), 5 (ASM) and 7, Type B messaging and SchedConnect, feeding a near-real-time single source of truth for schedule and flight data to **78 systems** across eleven downstream domains.

Production AI for operational safety: a fully AI-based callsign safety system (Neo4j, GAT, PyTorch Geometric) for an **NP-hard** Quadratic Assignment Problem, a real-time Early Warning System and an overflight cost engine.

Architecture underneath the domain: high-transaction-volume integration with **200+ integration points across 86 systems**, ~50M annual transactions and **99.8% availability**, 24/7; PMP-certified, rated **exceptional** in DDI Manager Ready.

## WORK EXPERIENCE

### Career Break and Professional Development | Istanbul | Sep 2024 – Present

- Planned break after leaving the Turkish Airlines Group: family time, structured upskilling (**Harvard's CS50 AI**, 12 Python projects; **AWS Certified AI Practitioner**, **PSPO I** and **AWS Certified Cloud Practitioner** in 2026, with foundation-level AWS Well-Architected Framework knowledge) and continued research on graph-based AI for aviation safety.

### Principal Solutions Architect | Turkish Technology | Istanbul, Turkiye | Jan 2023 – Sep 2024

*Official title: Distinguished Expert, the highest technical grade at Turkish Technology.*

- Cut reported ATC confusion events by **98%**, from **50 to 1 per season** (sustained since), on up to ~1,500 daily peak-season flights, replacing the 2010 heuristic season-start assignment with an in-house fully AI-based callsign safety system aligned with EUROCONTROL CSST.
- Shaped integration and technology strategy for next-generation aviation operations platforms as **NetLine Subject Matter Expert** for Turkish Airlines, SunExpress and AJet, and as SME in callsign safety, IOCC and operational control systems.
- Held schedule mismatches at **zero** across two carriers and two reservations systems during AJet's migration from TROYA, the group's reservations and inventory platform, to a separate third-party PSS, running SchedConnect-based synchronization, inter-carrier schedule distribution and schedule change management in a transition architecture.
- Reframed callsign assignment as an **NP-hard** Quadratic Assignment Problem and built a scalable machine learning solution on **Graph Neural Networks (GNN)**: **Graph Attention Networks (GAT)** on Neo4j with PyTorch Geometric and feature engineering over phonetic, visual and semantic similarity.
- Reduced monthly incidents from **8 to 1** in 10 months as owner of roadmaps and technical direction for **six mission-critical operational systems** (1,000 users, ~100 concurrent, 5M annual transactions, EUR 1M budget), sequencing change through **integration dependency management** and architectural constraints.
- Ran the NetLine/Ops replacement selection (**5 vendors, 3 shortlisted, 2 POCs**; a **EUR 1.5M** decision) and vendor evaluation of flight planning platforms (Jeppesen, CAE formerly FPM, Lido) through RFP/RFI, vendor management and business cases.
- Advised VP-level leadership on build-versus-buy, technology strategy and long-term R&D strategy.
- Established design authority and design reviews that brought architecture decisions under ISO 27001 compliance**, software quality management and enterprise architecture governance**, and supported regulatory audits and ICAO compliance.
- Led an **8-person cross-functional team** (project manager, 2 system analysts, 1 business analyst, 4 software engineers) that delivered **9 systems to production**, as architecture and design decision owner, with Scrum Master support as required.
- Defined the implementation approach, mentored engineers hands-on and cleared complex technical blockers.

**Senior Integration Architect** | **Turkish Technology** | **Istanbul, Turkiye** | **Jan 2021 – Jan 2023**

*Official title: Senior Expert, Integration Architect.*

- Eliminated dispatches into restricted airspace (**18** in the 12 months before go-live, **zero** since) by engineering a real-time Early Warning System across NetLine and FPM: **72** automated checks and alarms on planned routes for up to **~1,500 daily** peak-season dispatch legs.
- Enabled **1.5M** disrupted passengers a year to re-accommodate online, even those en route to the aircraft, by extending PSS/DCS integration with the in-house PSS TROYA to irregular operations (IROPS/IRROPS) and disruption management, propagating IOCC decisions to inventory, reservations and departure control as they happened.
- Drove the enterprise integration transformation: standardized **200+ integration points across 86 systems** (**~50M annual transactions**) under one integration governance and data governance model, with enterprise messaging and data streaming reference patterns adopted group-wide for interoperability, maintainability, observability and operational reliability.
- Shortened new-integration lead time from **one week to one day**, cut integration-related incidents by **99%** and retired **47 legacy point-to-point interfaces** by reusing the group-wide reference patterns.
- Achieved **99.8% availability** (planned maintenance included) and **zero unplanned downtime** on mission-critical platforms serving **500+ concurrent users**, leading cross-functional delivery management across internal and external integration platforms through production deployment under high availability, disaster recovery design and SLA/SLO management.
- Modernized the integration landscape onto an event-driven application and data integration platform on Apache Kafka (topic design, schema registry), IBM MQ and REST/SOAP APIs, sustaining **~50M annual transactions**.
- Applied microservices, enterprise integration patterns, API-first design and API lifecycle management, including naming conventions, API versioning and backward compatibility.
- Owned the solution architecture artefacts that kept decisions traceable to a maintained target state architecture: solution blueprints, architecture roadmaps, High-Level and Low-Level Design (HLD/LLD), Architecture Decision Records (ADR) and non-functional requirements (NFR).
- Raised release frequency from **1 to 4 a month** and **halved failed releases** by setting the estate's engineering practice: CI/CD architecture and DevOps release automation with Git-based (GitOps) configuration control, automated testing strategies, and performance and security testing gating releases.
- Ran **Scrum**-based agile delivery in **Jira** and **Confluence**: user stories with acceptance criteria, backlog management, sprint planning and Definition of Ready/Done.
- Caught **99%** of integration failures before users reported them and reduced mean time to detect from **one hour to one minute** with a real-time observability tool built across all live enterprise integrations.
- Based the tool on structured logging, correlation IDs and distributed tracing, and directed delivery of Grafana dashboards, metrics and alerting to support proactive issue detection.

**Lead Solutions Architect and Platform Owner** | **Turkish Airlines** | **Istanbul, Turkiye** | **Jun 2014 – Jan 2021**

*Official title: Project Manager, Flight Operations.*

- Designed and built **TKAODS** (Airline Operational DataStore) and evolved **OFPDS** (Operational Flight Plan DataStore, live since 2010), which together became the **single source of truth** for schedule and flight data.
- Took schedule and flight-change delivery from **next-day to near-real-time** for **78 consuming systems**, replacing **70 separate flight-data interfaces** by routing every change through both platforms, 24/7.
- Architected both as **Tier-0** integration layers in front of the Lufthansa Systems NetLine suite (NetLine/Sched, NetLine/Slot, NetLine/OPS & Base) and flight planning.
- Managed the OCC/IOCC's daily operational message and data flows, 24/7: movement control and aircraft rotation/tail assignment in NetLine/OPS, IATA Type B MVT and LDM messages, ACARS feeds, NOTAMs and crew management interfaces.
- Owned enterprise-wide flight plan and flight data distribution for planning, operational and post-flight periods as the single integration point for **eleven main downstream domains**: scheduling, crew, load control, PSS/DCS interfaces, cargo, maintenance, DWH, revenue and cost accounting, SAP finance, HR/payroll and regulatory reporting.
- Processed **~100K update records a day** (peak-season average) for up to **~1,500 flights a day**, 24/7 and near-real-time, across summer and winter seasons.
- Consolidated, transformed and enriched every update through source-to-target mappings, ETL/ELT pipelines and integration rules under data quality and metadata controls.
- Architected batch, near-real-time and real-time distribution patterns that absorbed **~22.5M** flight-data changes a year (~450K annual flights, ~50 changes per flight) before fan-out to large-scale distributed data processing and analytics.
- Set consumer-specific horizons, from a **rolling three-day window** to a **90-day window**.
- Fed operational changes (delays, cancellations) to passenger-notification systems, triggering **75M notifications a year**, and flight data to loyalty and disruption hotel systems, so all three ran on the same data as operations.
- Designed the enterprise operational data and integration architecture on the AODB (Airline Operational DataStore) pattern, standardizing flight, leg, schedule and operational flight-plan data airline-wide: logical data models, entities and relationships, canonical data structures, and the ESB pattern applied twice.

- Evolved both Tier-0 platforms for **ten years** (2014 – Sep 2024) as de facto product owner with final authority over feature prioritization and architectural evolution.
- Coordinated operational stakeholders and cross-team dependencies for two Tier-0 platforms, carrying releases through production under ITIL-aligned **Site Reliability Engineering (SRE)** practices and continuous monitoring.
- Drove technical direction, integration and platform modernization for airline operational systems with a **EUR 6.5M** total programme budget (scheduling, slot management, flight planning, IOCC) as lead architect with delivery accountability.
- Worked end to end: system and business analysis, software architecture and hands-on coding through release, operational handover and 24/7 production under ITIL.
- Managed NetLine's master and reference data (NetLine/Base) with the team: built and maintained the tables where users edit it and trained key users in master data management (MDM), keeping master data owned at its source.
- Built a reporting layer into both platforms on the same enriched data and fed it back into NetLine and the flight planning system, so enterprise reporting and operations shared one version of the data.
- Kept **850 NetLine users** working through the pandemic by coordinating, as NetLine product owner, the infrastructure teams' VMware solution.
- Received Turkish Airlines **achievement certificates** in 2017 for contributions to the Fuel Management Information System, the FPM & FE implementation and the OpsEye Diversion Manager module and its rollout.

### Integration Solution Design Lead | Turkish Airlines | Istanbul, Turkiye | Jan 2007 – Jun 2014

*Official title: Acting Team Leader.*

- Delivered **EUR 8M** in cumulative savings from 2012 to 2023 (**EUR 800K** in the first year) with a multi-currency overflight cost engine built on a data pipeline of country entry/exit timestamps and segment distances.
- Led a **6-person team** through the lifecycle of **100+ integrations**, from requirements and system analysis to design, development, testing and documentation, harmonizing project plans and tracking cross-functional milestones per project.
- Architected the schedule-to-PSS (TROYA) message integration under IATA SSIM Chapter 4 (SSM) and Chapter 5 (ASM) over Type B messaging, keeping inventory and reservations in step with cancellations, re-timings, equipment and routing changes, reinstated flights and disruption reason codes.
- Introduced SchedConnect in 2007 for inter-carrier schedule enrichment and distribution, establishing recurring schedule data exchange with **54 partner carriers** under IATA data standards.
- Headed solution design and delivery for flight operations, crew management system and flight-planning integrations across global aviation vendors, including IATA Type B MVT and LDM message flows.
- Built multithreaded real-time integration frameworks (2010) and contributed to an in-house Enterprise Service Bus (ESB) on Service-Oriented Architecture (SOA).
- Built **OFPDS** in 2010 to cut mainframe dependency: flight plans from the new planning system landed in Oracle and the data warehouse, easing SAP integration; the fuel efficiency team rated them far more accurate than the earlier Jeppesen feed.
- Developed the first heuristic Callsign Similarity & Assignment System (phonetic, visual and semantic similarity), delivered in 2010: the groundwork later rebuilt as the graph-based AI solution once reclassified as NP-hard.
- Commended in writing in 2010 by Turkish Airlines' **Chief Pilot and Head of Flight Operations** for flight operations projects, the callsign project first among them, and personally by the **General Manager and CEO**.

### Software and Integration Engineer | Turkish Airlines | Istanbul, Turkiye | Apr 2000 – Jan 2007

*Official title: Analyst Programmer.*

- Contributed to the first production implementation of the Lufthansa Systems NetLine suite across schedule planning, slot management, movement control and aircraft rotation/tail assignment, the platform foundation owned for two decades.
- Cut costs by **EUR 100K** a year by designing and building the timetable publication system from scratch: worldwide airline schedules and partner codeshare flights in, a PDF timetable for web download and print out, so the scheduling department began preparing the timetable books itself (Java, C, PL/SQL).
- Administered NetLine at application level with 24/7 availability responsibility (upgrades and patches on AIX, database schema upgrades) and built its bi-directional real-time interfaces to the IBM mainframe on IBM MQ.
- Drove the technical work of the first major legacy modernization from day one as an implementation team member, moving schedule and operations control off the IBM mainframe (ADABAS/Natural) onto Oracle on Unix (AIX) and NetLine, retiring mainframe-era technical debt.
- Designed and built **Flight Watch**, a 24/7 world map tracing every Turkish Airlines flight in the air with simulated real-time positions: architecture decisions, mainframe integration over IBM MQ and UDP multicast (Java, Oracle, ILOG JViews Maps).
- Built the 24/7 middleware interfaces between the mainframe and **SCORE**, the system the Slot Coordination Center used to handle airline slot requests and check actual arrival and departure times against declared schedules; also handled its upgrades and configuration (Java, MySQL).
- Ran application-level administration of the **ACARS** system, keeping air-to-ground and ground-to-air messaging over the SITA network available 24/7, including software upgrades and system configuration.

- Ran statistical analysis of historical data to determine the methodology for the project that calculated **average passenger and baggage weights**; the methodology was adopted for weight-and-balance calculations.

## TECHNICAL SKILLS (hands-on) & CORE COMPETENCIES

**Flight Operations Systems:** Lufthansa Systems NetLine suite (NetLine/Sched, NetLine/Slot, NetLine/OPS & Base, SchedConnect), operations control center (OCC/IOCC), operational and tactical flight planning (Jeppesen flight planning; FPM, now CAE Flight Planning Solutions; Lido), schedule and slot management, operational data distribution, irregular operations (IROPS) and disruption management, PSS/DCS, DWH, SAP and MRP interfaces.

**IATA and Industry Standards:** SSIM Chapter 7 (Presentation and Transfer of a Schedule Data Set); schedule-to-PSS distribution across three group carriers.

**Operational Data Architecture:** Authoritative flight-data model and single source of truth, Canonical Data Model, near-real-time distribution to downstream operational consumers, Enterprise Data Architecture, Data Modeling, Operational Data Store (ODS), Data Integration Architecture, Data Pipeline Architecture.

**Integration Architecture:** Event-Driven Architecture, API Architecture, Apache Kafka, IBM MQ, ESB, REST/SOAP APIs; GraphQL (design level).

**Analytics & Graph / Advanced Analytics and AI:** Neo4j and graph databases, PyTorch and PyTorch Geometric, MLOps, data quality and metadata management.

**Engineering Platforms & Languages:** Oracle DB, Docker and containerization, Linux (RHEL) and Unix administration (AIX, HP-UX), application administration on Windows Server; Languages: Python, Java, C, PL/SQL, shell scripting (Bash, KSH), YAML and JSON; Git and Jenkins; UML, object-oriented design and system design; multithreaded real-time and database-driven systems.

**Operational Resilience:** 24/7 mission-critical operations, high availability and disaster recovery, SLA/SLO management, incident prevention, incident analysis, performance tuning, operational continuity.

**Executive Leadership and Delivery:** Transformation governance, executive (C-level) stakeholder management, programme management, delivery governance and operational readiness, customer consulting and workshop facilitation.

## EDUCATION

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|---|---|---|---|
| • Ph.D. (coursework) | Business Administration (Finance) | Yeditepe University | Istanbul |
| • M.Sc. | Management Engineering | Istanbul Technical University | Istanbul |
| • B.Sc. | Computer Science Engineering | Istanbul University | Istanbul |

## LANGUAGES

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|---|---|---|
| • English | C1 | Full Professional |
| • German | A2 | Studying towards B1 |
| • Turkish | | Native |

## CERTIFICATIONS

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|---|---|---|
| • AWS Certified Cloud Practitioner (CLF-C02) | Amazon Web Services (AWS) | 2026 |
| • AWS Certified AI Practitioner (AIF-C01) | Amazon Web Services (AWS) | 2026 |
| • CS50's Introduction to Artificial Intelligence with Python (12 projects) | Harvard University (CS50) | 2026 |
| • Professional Scrum Product Owner I (PSPO I), Credential ID 1347372 | Scrum.org | 2026 |
| • AI and Cloud Foundations: OCI AI Foundations; Agentic AI Foundations; IBM SkillsBuild; PMI AI (CPMAI, GenAI) | Oracle, IBM, PMI | 2025 |
| • Docker Foundations Professional Certificate | Docker | 2025 |
| • Manager Ready (DDI) leadership simulation: exceptional rating | TAG Taurus Group | 2024 |
| • ITIL v3 Foundation, Candidate No. 990453483 | BCS, The Chartered Institute for IT | |
| • Project Management Professional (PMP), Credential ID 1225730 | PMI | |
| • Lufthansa Systems SchedConnect vendor training (two weeks) | Lufthansa Systems | |